

ELEC5305 Project Proposal

Teacher: Craig Jin

1 Project title

Environmental Sound Classification Using Audio Feature Extraction and Machine Learning

2 Student Information

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GitHub Project Link: <https://github.com/ysha0293/elec5305-project-530087661>

GitHub Pages Link: <https://ysha0293.github.io/elec5305-project-530087661/>

3 Project Overview

With the development of intelligent city, environment monitoring and the intelligent devices, Automatically identify different types of environmental sounds has more and more important application value. For example, in the city environment, the system can automatically identify different acoustic events such as car horns, sirens, drilling holes, dog barking and street music and so on by sounds.

The aim of this project is to build a environmental sound classification system by utilizing the audio features extraction and machine learning methods. The project will use the UrbanSound8K dataset as the main dataset. Firstly, we will preprocess the audio signal and transfer them to feature expression of suitable classification, then training the machine learning model, classifying the different types of environmental sounds.

This project firstly builds a simple baseline system, then further studies more efficient audio feature expression or classification methods. Finally, comparing different methods by using classification accuracy, fusion matrix or other category-level performance metrics.

4 Background and Motivation

Environmental Sound Classification is an important task in the audio signal processing. Compare to the voice, the environmental voice has a huge difference in some aspects of duration, frequency characteristics, background noise and recording conditions. Therefore, it is somewhat challenging to automatically identify different environmental sounds.

UrbanSound8K is a common dataset in the research of environmental sound. This dataset includes 8732 audio segments, which has 10 categories of city environmental sounds and provide

the pre-defined folds as model evaluation [1]. Therefore, this dataset is suitable for comparing the different audio classification methods.

The traditional method usually extracts the Mel-Frequency Cepstral Coefficients and other audio features first, then the sound classification is completed by using a machine learning classifier. Deep learning method also can be applied widely in this task. Piczak use the CNN to study the environmental sound classification [2]; Abdoli et al studied the method of end-to-end classification directly from audio signals using one-dimensional convolutional neural networks [3].

This project was chosen mainly because it combines audio signal processing with machine learning into a practical classification problem. At the same time, it has a strong connection with the contents involved in ELEC5305, such as sampling, spectral analysis, feature extraction and audio processing.

5 Proposed Methodology

5.1 Dataset and Data Split

This project will use the UrbanSound8K dataset and official metadata. The experiment will use the dataset provided the official folds conducting the model development and performance evaluation.

Training dataset is used for learning model parameters, validation dataset is used for model choosing and final test dataset is only used for final performance evaluation. By this way, it can avoid the problem of dataset leakage.

5.2 Audio Preprocessing

The audio files in the dataset has different length, sampling rate and channel configuration, therefore, unified preprocessing is required before feature extraction.

If it is dual channel audio, convert it to mono; If it is audio with different sampling rates, convert it to the same sampling rate; If it is not enough length, it needs to be padded when necessary.

5.3 Feature Extraction and Classification

Firstly, using the traditional audio feature extraction methods and machine learning classifier builds a baseline classification system. MFCC will be one of basic audio, because it can describe the spectral characteristics of the audio in a relatively compact form.

Once baseline, more work will be done on more effective representations or classifications of audio features. Previous studies have shown that systems based on CNN can learn effective audio feature representations [4]. The approach of knowledge transfer through existing Audio Representations has also been explored [5]. In addition, different training strategies such as between-class Learning have also been applied to deep sound recognition [6].

These researches will provide some potential direction for improving baseline later, but in the Proposal stage, the final method will not be restricted to a specific model

5.4 Training and Evaluation

All model training and parameter selection only use training data and validation data. If feature standardization is required, the relevant statistics will only be calculated based on the training data, and then the same transformation will be applied to the validation set and the test set.

Classification Accuracy will be the main assessment metric, and at the same, the confusion matrix analysis analyze which environmental sounds can be correctly classified and which categories are prone to confusion. In addition, Precision, Recall and F1-score can further analysis the different categories.

The entire project will mainly be implemented using MATLAB.

6 Expected Outcomes

This project is expected to eventually achieve a complete environmental sound classification system, including the audio pre-processing, feature extraction, model training, sound category prediction and performance evaluation.

Firstly, get a repeatable baseline category result, then improve the audio feature expression or category method and compare its result with baseline. Final experimental result will include the classification accuracy, confusion matrix and category-level performance indicators.

The MATLAB source code, experimental results and project documentation will be saved in the GitHub Repository. Meanwhile, the project progress and the final experimental results will be displayed using GitHub Pages.

7 Timeline

Week	Task
Weeks 1–2	Determine the project title and main objectives
Weeks 3–5	Literature review, dataset preparation, and data analysis
Weeks 6–7	Complete the audio pre-processing, feature extraction and baseline
Weeks 8–9	Research and implement an improved classification method
Weeks 10–11	Conduct the model optimization, compare the methods
Week 12	Sort out the experimental results, graph, GitHub Repository and GitHub Pages
Week 13	Complete the final report and project display

References

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